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A stacking-based deep learning ensemble for multi-class skin lesion classification

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Abstract

Context Skin cancer is a major health burden in the world, and the incidence continues to rise and when diagnosed late, death rates are high. Dermoscopic imaging has emerged as one of the most important non-invasive diagnostic methods for early diagnosis; nevertheless, the interpretation of findings is largely reliant on clinical skills, which is driving the creation of dependable computer-aided diagnostic systems.

Existing literature and limitations More recent developments in deep learning, particularly convolutional neural networks (CNNs), have demonstrated promising performance in automated skin lesion classification. However, current methods based on single CNN networks or probability-level ensemble schemes often suffer from severe class imbalance, inter-class visual similarity, and poor generalisation. Such constraints are particularly observable in multi-class systems where minority and malignant lesion categories are often misclassified.

Objective of this research The main objective of this research is to address the gaps in current CNN-based and soft-voting ensemble systems by creating a more robust and discriminatory model for classifying multi-class skin lesions.

Present research This paper proposes a stacking-based deep ensemble model combining DenseNet121, EfficientNet-B0, and MobileNetV2 as complementary base learners. Each network is trained to produce deep feature embeddings, which are fused at the feature level and fed into an XGBoost meta-classifier to model nonlinear feature interactions and allow the model to learn feature weights.

Findings and results The results of extensive experiments on the HAM10000 dataset, comprising 10,015 dermoscopic images of seven lesion types, demonstrate that the proposed model achieves test accuracy of 86.38%, macro F1-score of 0.7485, weighted F1-score of 0.8559, and macro-AUC of 0.9880. The stacking ensemble demonstrated competitive and balanced performance compared with single CNN models and traditional soft-voting ensembles. To validate the robustness and clinical reliability of the proposed framework, additional experiments, including ablation analysis, statistical significance testing, Grad-CAM explainability analysis, and meta-classifier comparison, were conducted.

Contributions The paper has three main contributions: (i) it presents a deep feature-level stacking framework that is effective in capturing complementary representations by different CNNs, (ii) it shows the effectiveness of gradient-boosted meta-learning

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to deal with class imbalance and inter-class ambiguity, and (iii) it presents in-depth experimental evidence of enhanced sensitivity to clinically important minority-classes including AKIEC and melanoma.

Conclusion The results demonstrate that deep feature fusion and gradient-boosted stacking can improve classification robustness and balanced performance of multi-class skin lesion classification. The proposed framework presents a clinically meaningful solution for computer-aided dermatology and provides directions for future research, where uncertainty-aware learning, cross-dataset generalisation, and lightweight implementation in the clinical setting should be considered.

Keywords Deep learning, Ensemble learning, Skin lesion classification, Dermoscopy, XGBoost, Stacking, Feature fusion

1 Introduction

1.1 Context

Skin cancer is one of the most widespread malignancies worldwide, whose incidence continues to increase rapidly. Early and accurate diagnosis is essential towards better patient survival and reducing treatment complexity. Non-invasive imaging techniques, such as dermoscopy, have greatly contributed to the visualisation of the subsurface skin structure, thereby increasing the accuracy of diagnosis in pigmented skin lesions. Nevertheless, interpretation of dermoscopic images still heavily depends on clinical expertise, which often contributes to inter-observer variability and discrepancy in diagnosis [1, 2]. These difficulties underline the need for effective, objective computer-aided diagnostic (CAD) systems that can help clinicians screen and make decisions early.

1.2 Existing literature and limitations

Recent developments in deep learning, especially convolutional neural networks (CNNs), have achieved great success in analysing medical images and detecting melanoma. Despite these successes, the vast majority of current methods rely on single-CNN designs, are inherently vulnerable to class imbalance, often overfit to limited training data, and frequently fail to distinguish among lesion types with fine visual distinctions [3]. Such constraints are particularly strong in the context of multi-class skin lesion classification, where extreme data imbalance and morphological overlap among lesion types severely reduce classification effectiveness, especially for minority and malignant classes such as melanoma and actinic keratosis [4].

Ensemble learning is one of the strategies that have been investigated to enhance robustness and generalisation by combining multiple models. Nevertheless, most ensemble methods in use today rely on probability-level fusion schemes, e.g., soft voting or weighted averaging, which assume that model competencies are fixed (or equal) and do not leverage more complementary feature representations [5, 6]. Therefore, these methods remain insufficient to adequately address the imbalance in class distributions and the complex inter-class interactions in dermoscopic image analysis.

Recently, transformer-based dermatology models and hybrid CNN-transformer frameworks have shown great potential in capturing the global contextual representations from dermoscopic images. But many transformer-based methods require more data and computational power and might be less applicable in resource-limited clinical settings. Conversely, stacking-based CNN ensemble aims to improve complementary

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feature learning, computational efficiency, and enhance sensitivity to the minority class while ensuring consistent performance across imbalanced dermoscopic datasets.

1.3 Objective of this research

The main objective of this study is to address the shortcomings of single CNN models and traditional ensemble methods by developing a powerful, discriminative deep learning model for multi-class classification of skin lesions. In particular, the paper improves classification performance and sensitivity toward minority and clinically critical lesion categories using feature-level model complementarity and adaptive meta-learning.

1.4 Present research

In order to ensure this goal, the paper proposes a stacking-based deep ensemble architecture, which incorporates three of the most recent CNN models, including DenseNet121, EfficientNet-B0, and MobileNetV2 as base learners. Rather than being based on probability outputs, deep feature embeddings are obtained as a result of each CNN Global Average Pooling layer and combined to create a single high-dimensional representation. The composite feature vector is given to an XGBoost meta-classifier which learns nonlinear feature relationships and adaptive model contributions to enable more accurate and generalized multi-class decisions [7–9]. The novelty of the proposed framework is that it combines multiple CNN backbones to obtain multi-source heterogeneous deep feature embeddings and applies XGBoost for adaptive nonlinear meta-learning, thereby enhancing multi-class skin lesion discrimination.

1.5 Results and implications

Based on the HAM10000 experimental results, the proposed stacking ensemble performs much better than single-CNN baselines and traditional soft-voting ensembles. The framework demonstrates improved overall accuracy and F1 scores, as well as improved recognition of minority and malignant lesion categories, such as AKIEC and melanoma. The findings suggest that deep feature-level fusion with gradient-boosted meta-learning can be instrumental in eliminating class imbalance and inter-class ambiguity, two major problems in automated dermoscopic image analysis.

1.6 Contributions

The following are the key contributions of this study:

1. It proposes a rich feature-level stacking ensemble, which combines complementary representations from several CNN architectures for multi-class skin lesion classification.
2. It shows the efficiency of XGBoost-based meta-learning in the modeling of nonlinear interactions of features and improving sensitivity for minority lesion classes.
3. It gives an in-depth experimental analysis of the HAM10000 dataset, pinpointing the clinical significance and strength of the suggested framework in contrast to single-model and probability-degree ensemble structures.

1.7 Structure of the paper

The remainder of this paper is organized as follows. Section 2 reviews related work on deep learning and ensemble-based skin lesion classification. Section 3 describes the

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proposed methodology, including dataset preparation, base learner training, and the stacking strategy. Experimental results and performance analysis are given in Sect. 4. Section 5 discusses the findings and their implications for clinical practice, and Sect. 6 presents the conclusions and future research directions.

2 Related work and its limitations

2.1 Research scope and key issues

Automated skin lesion classification based on dermoscopic images is now a significant field of research due to the rising incidence of skin cancer and the need for a system that can confidently assist in diagnostic decision-making using computers. Methodologically, major challenges in this field include severe class imbalance, high intra-class visual similarity, and the inability to achieve strong generalisation in multi-class environments. These difficulties have inspired extensive research on deep learning-based models, especially convolutional neural networks (CNNs), in skin lesion analysis.

2.2 Review of existing literature and limitations

Recent publications in high-impact journals have shown that CNN-based methods can achieve high performance on the dermoscopic image classification task, particularly with the help of transfer learning and data augmentation. It has been demonstrated in a number of works that, properly fine-tuned CNN architectures and transfer-learning strategies greatly outperform conventional machine learning methods on standard benchmark datasets like HAM10000 [10–12]. Specifically, architectures built on the concepts of DenseNet, EfficientNet, and ResNet have become popular due to their strong feature detection capabilities and effective training methods.

Segmentation, augmentation, and classification of hybrid deep learning pipelines have also been investigated to enhance diagnostic accuracy [11]. EfficientNet scaled versions are reported to improve melanoma detection based on compound scaling strategies [13]. Although these approaches achieve better results, they primarily use single-model architectures that are vulnerable to data imbalance and sensitive to morphological overlap across lesion classes, which limits their performance in multi-class classification.

To solve these problems, ensemble learning methods have been explored more and more. According to studies, multiple CNN models have been shown to increase performance when combined via soft voting or weighted averaging schemes [10, 14]. Though these ensembles are more robust than individual models, they assume equal or fixed competencies of the base learners and operate at the probability level, thus not taking advantage of deeper levels of complementary feature representations. As a result, their adaptive flexibility in focusing on discriminative characteristics across lesion categories is limited.

More sophisticated optimization-based ensemble methods have also been suggested. For example, arithmetic optimisation and deep transfer learning have been applied to enhance melanoma classification, in which the final decision is made by weighted voting across multiple feature extractors [15]. They achieve more successful tuning but rely on pre-existing fusion rules rather than learning the optimal interaction between features, and so cannot flexibly apply to more complex multi-class classification problems.

In general, the literature review supports the use of CNNs, transfer learning, and ensemble learning in the classification of skin lesions. However, most prior approaches

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focus on probability-level fusion and they do not support learned integration of deep feature embeddings using a dedicated meta-classifier. Consequently, the possibility of adaptive fusion and nonlinear interaction modelling of deep feature representations has not been thoroughly studied, especially for minorities and clinically significant lesion types.

The proposed DL + ML pipeline has practical advantages over end-to-end deep ensemble architectures (e.g., simpler optimisation, modularised features, better handling of diverse deep representations). Ensemble models that adapt end-to-end can provide a joint feature adaptation, but they can be very expensive in terms of computational resources and training complexity. The proposed feature-level stacking framework, on the other hand, provides independent optimisation of the backbone and nonlinear decision modelling using XGBoost. Fixed feature embeddings might, however, prevent full end-to-end feature refinement. The proposed heterogeneous feature-level stacking framework achieved competitive classification accuracy, but end-to-end optimisation approaches can achieve more effective feature adaptation between the backbone feature extractor and the meta-level decision learner.

Recent hybrid and fusion-based deep learning techniques have been shown to achieve better classification performance by complementing the representation learning. The fusion of conventional descriptors with deep CNN features, feature fusion strategies, dimensionality reduction methods, and traditional machine learning classifiers resulted in better discriminative capability in medical and multi-class image classification tasks [16, 17]. Such methods have enhanced classification robustness, but they also require extensive conventional feature engineering, feature selection methods, or the use of traditional classifiers, e.g., SVM [16, 17]. Moreover, ensemble and hybrid deep learning methods have recently emerged, combining complementary feature extraction and adaptive classification to enhance the accuracy of skin lesion discrimination [18]. On the contrary, the proposed approach uses heterogeneous feature representations extracted from various CNN backbones and an XGBoost-based stacking ensemble to model decisions adaptively and non-linearly, leading to better sensitivity towards minorities and stronger discrimination among multiple skin lesion classes.

2.3 Motivation and objectives of the present research

The drawbacks identified in previous research motivated the need for a more adaptable and discriminative ensemble framework that goes beyond traditional soft-voting strategies. To this end, the current study proposes a stacking-based deep ensemble model that jointly leverages complementary deep feature representations from a series of CNN architectures and optimises their combination via gradient-boosted meta-learning. The proposed method aims at enhancing the classification accuracy, robustness, and sensitivity in multi-class dermoscopic image classification by learning nonlinear feature interactions at the ensemble level to predict minority lesion classes.

3 Methodology

3.1 Framework overview

The proposed stacking-based ensemble model consists of three major stages:

a. Preprocessing and augmentation of data,
b. Training by transfer learning with three CNN base learners, and

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c. The combination of these features is achieved using the XGBoost meta-classifier to perform the classification task.

The general layout of the planned framework can be explained by the need to address class imbalance and intra-class ambiguity in the context of multi-class classification of dermoscopic images. Specifically, feature-level stacking is preferred over decision-level fusion because the former maintains more semantic information across single CNN representations and allows model complementarities to be learned adaptively by a meta-classifier.

To start with, dermoscopic images will be preprocessed by using resizing, normalization, and class-balanced augmentation. These steps reduce imaging artefacts caused by illumination, imaging angle, and lesion shape variations, and they also add to the diversity of the data set. These preprocessing operations improve optimisation stability and enhance the robustness of the learned representation against real-world clinical imaging variations.

The second stage involves fine-tuning three complementary CNN models, including DenseNet121, EfficientNet-B0, and MobileNetV2, is performed on the basis of transfer learning. Each network independently develops hierarchical and discriminative representations of melanocytic structure. The classification layers are removed, and the high-level embeddings are obtained out of the GlobalAveragePooling2D layer of each CNN. These three architectures were chosen due to their architectural diversity that facilitates heterogeneous feature learning that is a prerequisite of successful stacking-based ensemble learning.

The final stage performs feature-level stacking using deep fusion embeddings. The feature vectors extracted from the three base networks are fused into a unified representation capturing different dermoscopic characteristics. This integrated feature representation is learned using an XGBoost meta-learner, which models nonlinear interactions across feature sets and optimises correct decision boundaries in multi-class classification. Figure 1 represents the general architecture.

3.2 Dataset description

HAM10000 dataset is chosen because of its clinical significance, class heterogeneity, and natural imbalance between classes, which makes it an appropriate dataset to test the resilience of the multi-class classification models. This study uses the HAM10000 ("Human Against Machine") dataset, a popular benchmark for dermoscopic image analysis [19]. It contains 10,015 high quality dermatoscopic images gathered from various clinical sources and labeled by specialist dermatologists. The sample includes seven categories of clinically significant lesions, which include Melanoma (MEL), Melanocytic Nevus (NV), Basal Cell Carcinoma (BCC), Actinic Keratoses/Intraepithelial Carcinoma (AKIEC), Benign Keratosis (BKL), Dermatofibroma (DF), and Vascular Lesions (VASC). All the images are resized to 224×224 and scaled to the range [0,1] to have the same input. The dataset is split into training (7007 images), validation (1499 images), and testing (1509 images), with 70, 15, and 15, respectively. Such stratified division guarantees impartial assessment without any distortion in the class distributions in all subsets. Figure 2 shows the seven different class samples from the HAM10000 dataset.

Large scale data augmentation is employed in training to improve generalization and alleviate the class imbalance that is inherent in HAM10000. The augmentation pipeline

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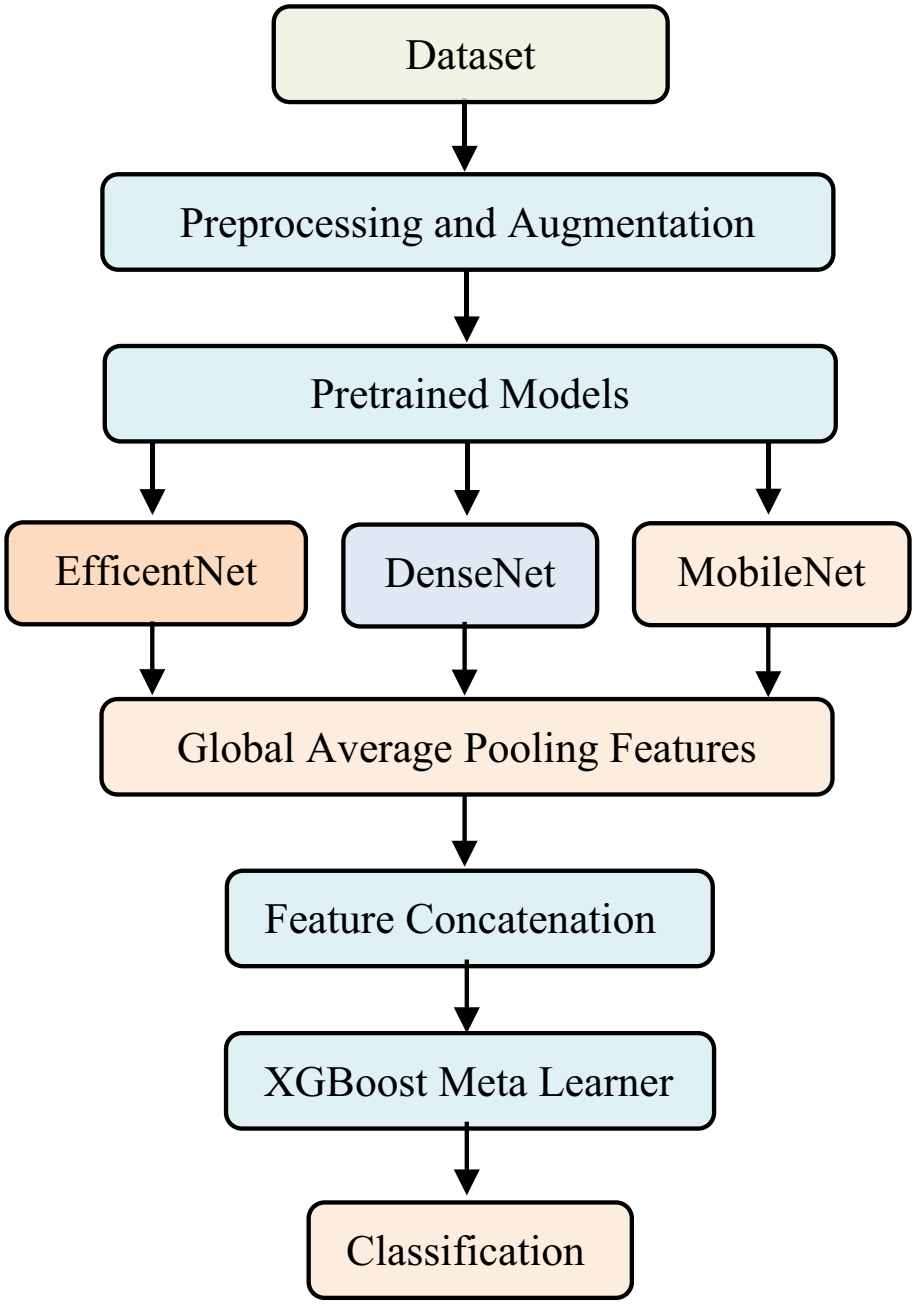


Fig. 1 Conceptual workflow of the proposed stacking ensemble for skin lesion classification

includes random rotations (within 15 degrees), zoom distortions, brightness distortions, horizontal and vertical flips and minor spatial distortions. These types of operations reproduce realistic changes in imaging variation observed in clinical practice, and the model becomes more robust to morphological variation in lesions.

3.3 Base learners

The base learners are selected based on complementary architectural characteristics and proven effectiveness in learning medical image representations. Three CNN-based architectures were selected as base learners, as they have complementary design

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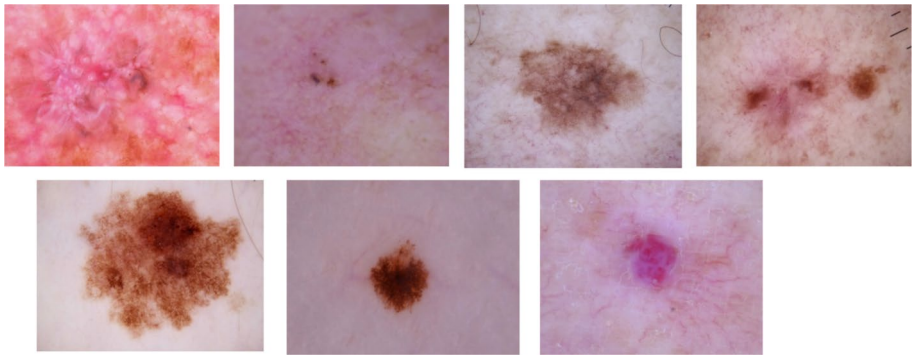


Fig. 2 Sample Dermoscopic images from the HAM10000 dataset

principles and features. All the models were trained using ImageNet-pretrained weights and fine-tuned on the HAM10000 training set with the Adam optimiser (learning rate 1×10^{-4}) and the EarlyStopping and ReduceLROnPlateau callbacks to avoid overfitting and ensure uniform convergence.

DenseNet121 consists of densely connected convolutional layers; hence, each convolutional layer takes features from all previous layers. The benefits of such architecture include a great deal of feature reuse, larger gradient propagation, and more richness of representation [20]. The strong density of connectivity allows the model to capture fine-grained structures of dermoscopic features which can be used to distinguish similar appearing lesions. After fine-tuning on HAM10000, DenseNet121 achieved a validation accuracy of 86.18% and demonstrated strong representation learning capacity [21]. It has a high level of connectivity, enabling the propagation of fine-grained features, which is especially valuable for differentiating lesion classes that are visually similar.

EfficientNet-B0 is a compound-scaling architecture, which achieves a balance between network depth, width, and resolution of inputs. It is highly accurate and computationally efficient [22]. EfficientNet-B0 demonstrated fluctuating training behaviour but achieved a strong validation accuracy of 80.27%, which demonstrates competitive learning capability for dermoscopic image classification. However, EfficientNet-B0 adds useful complementary feature representations thereby increasing ensemble diversity, another key factor in the performance of stacking-based meta-learning [23]. This complementary behavior improves ensemble diversity and contributes to better generalization in stacking-based meta-learning.

MobileNetV2 is a small but resource-efficient architecture. It has inverted residual blocks with linear bottlenecks and depth-wise separable convolutions, which are easy to compute and retain representational capacity [24]. MobileNetV2 converged smoothly and has a validation accuracy of 82.86%, indicating that it can learn compact but discriminative features [25]. Its lightweight composition adds compact yet discriminating features, enhancing the strength of the ensemble.

3.4 Deep feature extraction and stacking ensemble

Rather than using the softmax probability outputs, we make use of a deep feature-level fusion approach to build the stacking ensemble. Feature embeddings from the GlobalAveragePooling2D layer of every trained CNN base model yield small but highly discriminative representations of dermoscopic image features [26, 27]. To be more precise, the DenseNet121 model produces feature vectors with dimensions of 1024, whereas the

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EfficientNet-B0 and MobileNetV2 models produce feature vectors with dimensions of 1280. These feature vectors are obtained and merged to form 3584-dimensional descriptors of each input image, which capture complementary visual patterns learned by the three architectures. The final fused feature representation was a 3584-dimensional deep feature, obtained by concatenating DenseNet121 embedding (1024), EfficientNetB0 embedding (1280), and MobileNetV2 embedding (1280), which provided a heterogeneous, multi-scale lesion representation.

Let $f_i^{(k)} \in \mathbb{R}^{d_k}$ represents the deep feature of the CNN, of the th image at the kth layer. The final stacked feature representation can be defined as;

$$F_i = \left[f_i^{(1)} \middle| f_i^{(2)} \middle| f_i^{(3)} \right]$$

The combined feature representation was fed into an XGBoost classifier configured as follows: trees = 300, max depth = 6, learning ra = 0.05, and subsample = 0.8. XGBoost is also chosen due to its nonlinear interaction among features and adaptable decision borders, which cannot be realized in the case of fixed averaging and voting-based fusion approaches. XGBoost is most suitable in a stacking meta-learner, since it can be used to model nonlinear functions and interactions between features. XGBoost is a gradient boosting algorithm that has performed well on tabular and high-dimensional data, learning nonlinear decision boundaries and adaptive feature interactions that cannot be learned by simple averaging [28].

XGBoost parameters were determined empirically after several validation experiments, aiming to balance nonlinear learning ability, model complexity, and overfitting. Likewise, CNN training parameters like Adam optimization, learning rate scheduling, dropout regularization, early stopping, and class-weight balancing were chosen to ensure the convergence stability and sensitivity to minority-classes when severe class imbalance occurs.

3.5 Computational complexity and deployment considerations

The proposed stacking framework enhances classification robustness but introduces additional computational complexity due to the parallel use of multiple CNN backbones and feature concatenation. DenseNet121 offers solid discriminative representation learning and consumes more memory, while MobileNetV2 is more compact for lightweight scenarios. EfficientNet-B0 is a balance between performance and parameter efficiency. The proposed framework can still be applicable to offline clinical decision-support applications, but for real-time deployment in low-resource scenarios, model compression and light-weight optimization techniques will need to be explored in the future.

4 Results

The HAM10000 test set, consisting of 1509 dermoscopic images, was used to test the proposed stacking-based ensemble. The three CNN base learners were tuned separately (DenseNet121, EfficientNet-B0, and MobileNetV2) and then combined into an ensemble to ensure representational diversity. The validation accuracies of 86.18%, 80.27% and 82.86% suggest that all networks learn complementary lesion characteristics that are important for successful ensemble learning.

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Table 1 Performance of the proposed stacking ensemble model

Metric	Value
Test accuracy	86.38%
Macro precision	0.8390
Macro recall	0.6996
Macro F1-score	0.7485
Weighted F1-score	0.8559
Macro-AUC	0.9880

Table 2 Class-wise performance of proposed stacking ensemble model

Class	Precision	Recall	F1-score
AKIEC	0.8696	0.4000	0.5479
BCC	0.7375	0.7564	0.7468
BKL	0.7349	0.7349	0.7349
DF	0.9091	0.5556	0.6897
MEL	0.7680	0.5714	0.6553
NV	0.9018	0.9701	0.9347
VASC	0.9524	0.9091	0.9302

The XGBoost meta-learner was used to make the final classification based on concatenated deep feature embeddings of the base models. Table 1 reports that the proposed stacking ensemble had a test accuracy of 86.38% with a macro precision of 0.8390, macro recall of 0.6996 and macro F1-score of 0.7485. Moreover, the macro-AUC value of 0.9880 indicates that the seven lesion categories have high separability. These findings verify that the problem of multi-class skin lesion classification can be solved effectively using the combination of feature-level stacking and gradient-boosted meta-learning.

In terms of problem solving, the results obtained show that the proposed framework is effective in solving major problems like class imbalance and inter-class visual similarity. The stacking ensemble learns more discriminative decision boundaries on top of fused deep feature representations to be more sensitive to clinically significant lesion types whilst preserving overall classification strength, which is important to computer-aided dermatology systems.

4.1 Class-wise performance analysis

Table 2 shows the performance of the proposed stacking ensemble in terms of the classes. The findings reveal that both majority and minority classes show strong and balanced performance. Common types of lesions, such as NV, BCC, and VASC, show high precision and recall, indicating reliable classification of these lesions. However, more importantly, a better recall in the minority and malignant classes like AKIEC and melanoma is indicative that the model is more sensitive to the clinically critical cases.

The relatively low recall in the case of DF can be explained by its low representation in the dataset and its visual similarity with benign lesion types. This problem has been widely reported in earlier research of dermoscopic image analysis and highlights the intrinsic complexity of minority-class classification in strongly biased medical data sets. However, the proposed ensemble performs competitively even under these demanding conditions, indicating the power of the learned feature representations.

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Table 3 Class-wise performance of the soft-voting ensemble

Class	Precision	Recall	F1-score
AKIEC	0.7222	0.2600	0.3824
BCC	0.7115	0.7179	0.7147
BKL	0.6503	0.6928	0.6709
DF	0.8333	0.5556	0.6667
MEL	0.6970	0.5476	0.6133
NV	0.8940	0.9651	0.9282
VASC	0.9474	0.8182	0.8780

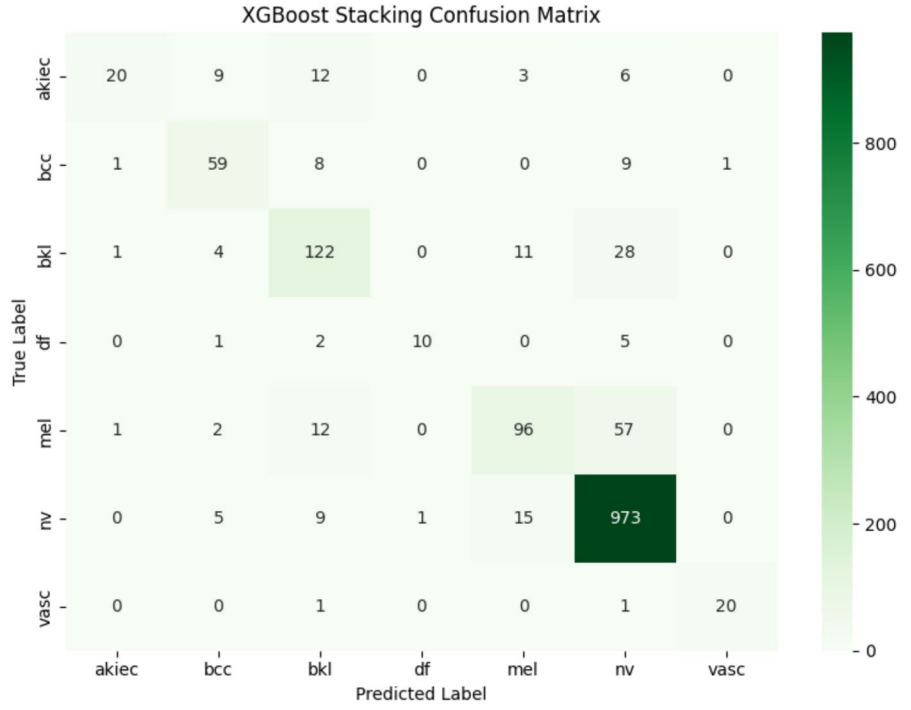


Fig. 3 Confusion matrix of the proposed XGBoost-based stacking ensemble on the HAM10000 test dataset

4.2 Comparison with soft-voting ensemble

To further evaluate the efficiency of feature-level stacking, the suggested approach was compared with a traditional soft-voting ensemble, as summarised in Table 3. Although the soft-voting method provides good overall performance across several lesion classes, it has lower recall rates on minority lesion categories like AKIEC and melanoma.

This performance difference is expected, since probability-level fusion assumes fixed competence among base learners, and no manipulation of intermediate feature representations. In comparison, the suggested stacking method enables the XGBoost meta-learner to capture nonlinear interactions among deep features, resulting in better generalization and higher sensitivity to the minority-class. The results indicate that learned feature fusion is better than fixed-point voting strategies.

4.3 Confusion matrix and its analysis

The confusion matrix of the proposed stacking ensemble is shown in Fig. 3. The visual confirmation of high true-positive rates in the majority of lesion types is made by the matrix (NV, BCC, and VASC). The proposed framework also resulted in an improved

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recognition of clinically relevant malignant lesion categories, including recognition of melanoma and AKIEC, as compared to traditional soft voting methods.

The misclassifications left behind are mostly those between similar classes like MEL and NV. This confusion pattern is typical of the inherent morphological overlap in dermoscopic diagnosis, and it has been reported in large numbers in the dermatology literature. Such misclassifications are thus indicative of the inherent clinical difficulty of the multi-class skin lesion discrimination task, rather than of random prediction behaviour.

4.4 ROC analysis

Figure 4 presents the ROC curve comparison between the DenseNet121 backbone, the soft-voting ensemble, and the proposed XGBoost-based stacking framework. The proposed stacking ensemble achieved the highest macro-AUC value of 0.9880, outperforming both the standalone DenseNet121 model and the soft-voting ensemble. The ROC analysis demonstrates that the proposed framework learns stable and well-separated decision boundaries for multi-class skin lesion classification. Furthermore, the consistently high macro-AUC values indicate strong discriminative capability despite the severe class imbalance present in the HAM10000 dataset.

4.5 Meta-classifier comparison

Further experiments were performed on the same stacked deep representations using Logistic Regression, Random Forest, and Support Vector Machine (SVM) classifiers, as shown in Table 4, to support the selection of XGBoost as the meta-classifier. Although SVM achieved slightly higher classification accuracy, XGBoost demonstrated competitive and balanced performance under class-imbalanced conditions and better captured complex relationships within the fused deep embeddings. In addition, XGBoost

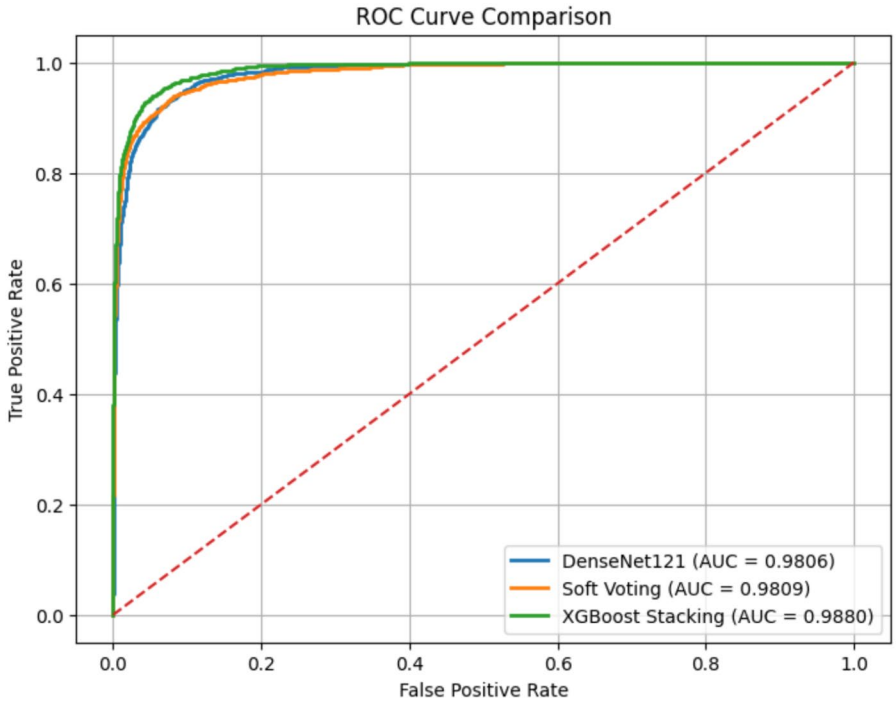


Fig. 4 ROC curve comparison between DenseNet121, soft-voting ensemble, and the proposed XGBoost stacking framework on the HAM10000 dataset

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Table 4 Performance of the Meta-classifiers

Meta-classifier	Accuracy (%)	Macro-F1
Logistic Regression	85.04	0.7823
Random forest	84.25	0.5905
SVM	86.50	0.7663
XGBoost	86.38	0.7485

Table 5 Ablation study showing the contribution of individual backbone networks and their feature-level combinations

Configuration	Accuracy (%)	Macro-F1
DenseNet121	86.18	0.7522
EfficientNetB0	80.27	0.6265
MobileNetV2	82.86	0.6350
Dense + Efficient	86.51	0.7414
Dense + Mobile	86.51	0.7366
Efficient + Mobile	83.99	0.6813
All three models	86.11	0.7445

provided more consistent discrimination of minority lesion categories and improved interpretability of feature contributions. Therefore, XGBoost was selected as the final meta-classifier for the proposed framework.

4.6 Ablation study

To analyze the contribution of each CNN backbone and their combination in the stacking framework, an ablation study was carried out as shown in Table 5. The analysis showed that DenseNet121 on its own had the best performance, and that EfficientNet-B0 and MobileNetV2 provided complementary feature representations that improved ensemble diversity and minority-class discrimination. The three-model fusion provided balanced performance across lesion categories while improving feature diversity.

The ablation analysis demonstrates that DenseNet121 individually achieved the strongest standalone performance among all backbone networks, highlighting its superior feature extraction capability for dermoscopic image classification. Although the fusion of multiple backbone networks increased representation diversity and improved discrimination of underrepresented lesion categories, the combination of all three models provided a balanced trade-off between overall classification accuracy and macro-F1 performance. These findings confirm that heterogeneous feature fusion contributes positively to multi-class lesion discrimination.

4.7 Statistical significance analysis

To confirm the effectiveness of the proposed stacking-based framework relative to baseline ensemble strategies, a statistical significance analysis was conducted comparing the soft-voting ensemble with the proposed XGBoost-based stacking framework. A paired t-test was used to analyze prediction results from the test set of HAM10000. The p-value obtained, 0.002998, is well below the conventional 5% threshold, suggesting that the performance improvement obtained by the proposed framework is not due to random variation but rather to statistical significance.

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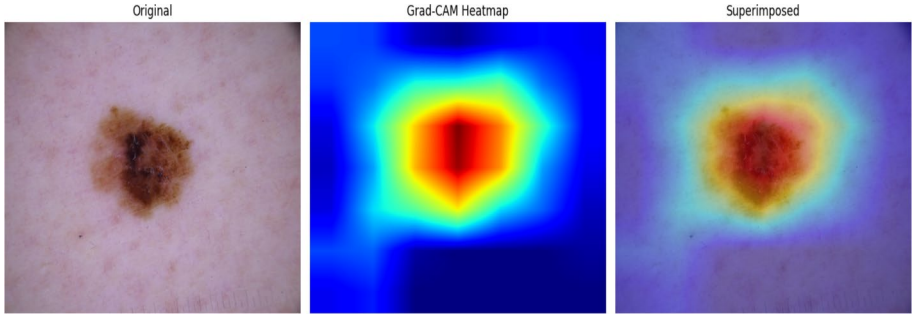


Fig. 5 Grad-CAM visualization for melanoma lesion classification

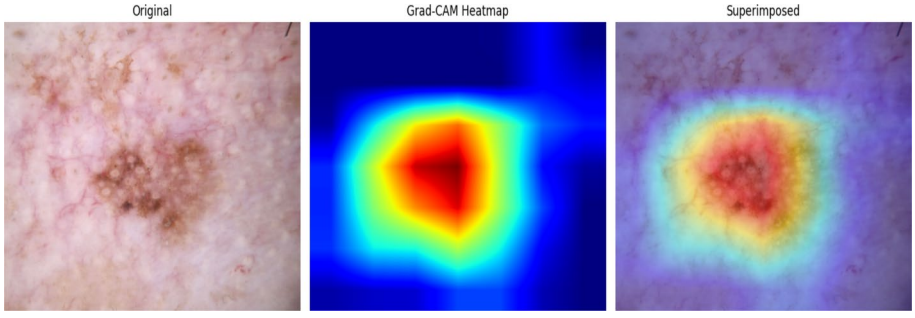


Fig. 6 Grad-CAM visualization for AKIEC lesion classification

Table 6 Comparison of the proposed model with state-of-the-art literature

Method	Accuracy	Precision	Recall	AUC/other metrics
MobileNet-based skin lesion analyser [29]	83.1%	0.89	0.83	Top-2: 91.36% Top-3: 95.34%
ESRGAN enhanced CNN Model [30]	86%	0.84	0.86	F1 Score 0.86
Skinsight deep CNN model [31]	90.18%	0.91	0.55	Dice: 90.80% Jaccard: 83.37%
Proposed stacked deep ensemble	86.38%	0.8390(Macro)	0.6996(Macro)	AUC 0.9880

4.8 Explainability analysis using grad-CAM

For better clinical interpretability and explainability, the proposed framework was used with Gradient-weighted Class Activation Mapping (Grad-CAM) to visualise the learned discriminative lesion regions by the DenseNet121 backbone network, as shown in Figs. 5 and 6. The heatmaps generated reveal that the proposed framework is mainly dealing with clinically relevant lesion structures and high-attention pathological regions without focusing on background artefacts. The model focused on asymmetric pigmented areas and irregular lesion margins in melanoma samples, while the activation maps in AKIEC samples included localized keratotic regions and abnormal textured areas. The results show that the proposed ensemble can identify meaningful dermatological representations that are similar to clinically relevant visual patterns.

4.9 Comparison with the state-of-the-art literature

Table 6 compares the performance of the proposed approach with state-of-the-art methods that were reported in the literature to contextualize its effectiveness. Although the existing models are reported to have slightly higher precision, they tend to have significantly lower recall, especially for malignant lesion categories. An example is the

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SkinSight model with a high precision but limited sensitivity, which could decrease its reliability when used in clinical screening.

Comparatively, the proposed stacking ensemble has a more balanced set of quality metrics, with accuracy and sensitivity. The increased recall for malignant and minority lesion classes, together with the high AUC values, is consistent with previous findings in medical image analysis literature that emphasizes high recall and robustness rather than high accuracy. These findings indicate that the proposed framework provides competitive and balanced performance across a variety of tasks compared to current CNN-based and ensemble-based frameworks.

4.10 Summary of experimental findings

In general, the findings demonstrate that deep feature-level stacking with gradient-boosted meta-learning offers a promising and clinically significant improvement in multi-class classification of skin lesions. The proposed method is more sensitive to minority and malignant lesion categories, more robust under class-imbalanced conditions, and achieves better performance than probability-level ensemble methods. The results emphasize the significance of adaptive feature fusion and meta-learning in developing dependable computer-aided diagnostic and offer a solid basis for further development of the concepts including cross-dataset validation, uncertainty-conscious prediction, and real-time clinical application.

5 Discussion

The experimental findings demonstrate that feature-level stacking effectively combines complementary representations from multiple CNN architectures for multi-class skin lesion classification. Compared with conventional soft-voting ensembles, the proposed framework demonstrated stronger discrimination, particularly for clinically important lesion categories.

The meta-classifier comparison showed that although SVM achieved slightly higher classification accuracy, XGBoost offered a better balance between generalization ability, nonlinear feature interaction learning, and robustness in an imbalanced medical imaging scenario. Moreover, XGBoost provides adaptive gradient-boosting decision modeling suitable for heterogeneous deep feature fusion extracted from multiple CNN backbones.

The Grad-CAM visualizations showed that the proposed framework focused on the clinically relevant regions of the lesion rather than background artifacts, which further improves interpretability and increases confidence in the model predictions. For melanoma lesions, the activation maps were more focused on the pigmented areas, which were generally asymmetric, and irregular boundaries of the lesions, while the activation maps for AKIEC were more concentrated on the localized keratotic and abnormal texture patterns.

Although the classification results obtained on HAM10000 are good, there are some drawbacks. The dataset is imbalanced and reflects only a limited clinical spectrum, which could affect generalizability to other clinical settings. Moreover, the proposed framework is based exclusively on dermoscopic information in an image rather than on clinical information such as patient age, lesion location, or patient history. Moreover, training and inference of multiple CNN backbones are also computationally complex. Although the proposed framework achieved strong performance, end-to-end

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optimization may further improve feature adaptation, a promising avenue for further research. Future research will be oriented towards external cross-dataset validation, domain adaptation, uncertainty-aware prediction, multimodal learning and lightweight optimization strategies for real-time clinical deployment.

6 Conclusion

In this study, a feature-level stacking ensemble method for multi-class skin lesion classification was introduced by stacking DenseNet121, EfficientNet-B0, and MobileNetV2 deep feature representations and using the XGBoost meta-classifier. The proposed framework achieved 86.38% accuracy on the HAM10000 dataset, with macro-F1 of 0.7485, weighted F1 of 0.8559, and macro-AUC of 0.9880, demonstrating strong discriminative power in dermoscopic lesion analysis. The effectiveness of the proposed stacking strategy for heterogeneous deep feature fusion was confirmed through comparative experiments with soft voting ensembles and multiple meta-classifiers.

The robustness and clinical relevance of the proposed framework were further confirmed through additional experiments such as ablation analysis, statistical significance testing, Grad-CAM explainability analysis, and meta-classifier comparison. The Grad-CAM visualisations confirmed that the model primarily highlighted clinically relevant areas of the lesions, thereby enhancing interpretability and diagnostic confidence. The proposed framework demonstrated strong internal performance, but further external validation across diverse datasets is still needed, along with lightweight optimisation, multimodal integration into clinical workflows, and improved generalisation.

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Author contributions
V.C.H. conceived the research idea, designed the methodology, implemented the models, performed the experiments, and wrote the initial draft of the manuscript. S.S. contributed to the literature review, data analysis, and manuscript editing. W.J. assisted in methodology refinement and provided technical guidance on deep learning architectures. S.M.N.I. supervised the research work, contributed to result interpretation, and critically reviewed and revised the manuscript. K.C. assisted in validation, critical revision of the manuscript, and overall technical improvements. All authors reviewed and approved the final manuscript.

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Data availability
The dataset in this research paper (HAM10000) is publicly accessible, and it may be downloaded using its official repository. The code used for model training and evaluation is available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable. This study used a publicly available dermoscopic image dataset (HAM10000) and did not involve human participants, clinical trials, or animal experiments conducted by the authors.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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